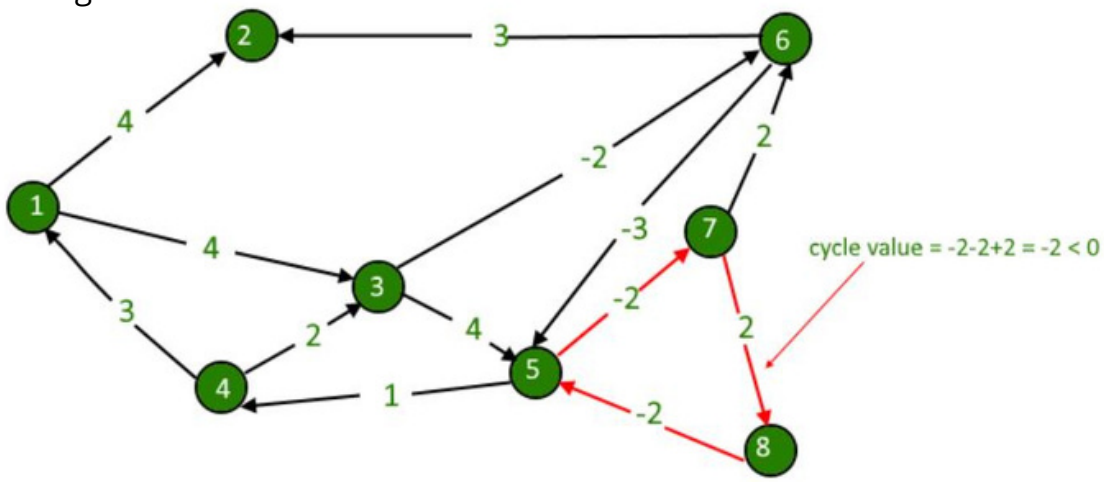
Experiment No. 7

AIM: Implement all pairs shortest path- Floyd Warshall Algorithm.

THEORY:

- Floyd Warshall Algorithm is an example of dynamic programming
- approach.
- It computes the shortest path between every pair of vertices of the given
- graph.
- It is used to solve All Pairs Shortest Path Problem.
  It is best suited for dense graphs.
- This is because its complexity depends only on the number of vertices in
- the given graph.
- Algorithm works for graphs with negative edge weights also.
  Negative cycles are not allowed.
  A negative cycle is one in which the overall sum of the cycle comes negative.



For every pair (i, j) of the source and destination vertices respectively, there are two possible cases.

• k is not an intermediate vertex in shortest path from i to j. we keep the value of dist[i][j] as it is.

• k is an intermediate vertex in shortest path from i to j.

We update the value of dist[i][j] as dist[i][k] + dist[k][j]

if dist[i][j] > dist[i][k] + dist[k][j]

Analysis of algorithms

ALGORITHM:

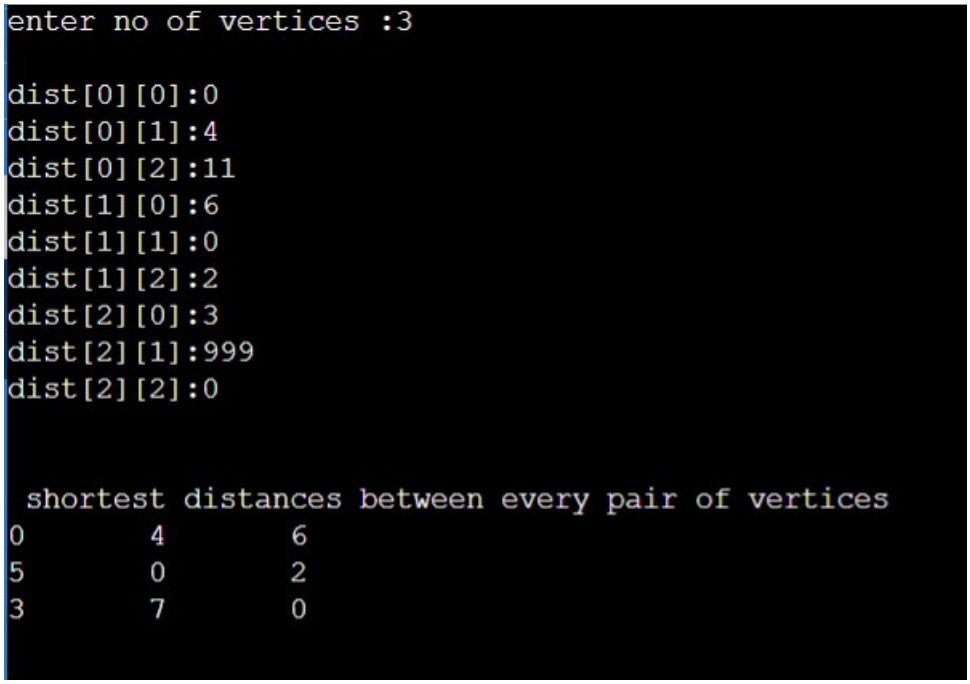
```
floyd_warshall(int* W, int n) {
  for (int k = 0; k < n; k++) {
    for (int i = 0; i < n; i++) {
      for (int j = 0; j < n; j++) {
        W[i][j] = min(W[i][j], W[i][k] + W[k][j]);
      }
    }
  }
}
```

SAMPLE CODE:

```
#include<stdio.h>
int i, j, k,n,dist[10][10];
void floydWarshell ()
{
 for (k = 0; k < n; k++)
  for (i = 0; i < n; i++)
   for (j = 0; j < n; j++)
    if (dist[i][k] + dist[k][j] < dist[i][j])
     dist[i][j] = dist[i][k] + dist[k][j];
}
int main()
{
  int i,j;
  printf("enter no of vertices :");
  scanf("%d",&n);
  printf("\n");
  for(i=0;i<n;i++)
  for(j=0;j<n;j++)
   {
    printf("dist[%d][%d]:",i,j);
    scanf("%d",&dist[i][j]);
   }
 floydWarshell();
 printf (" \n\n shortest distances between every pair of vertices \n");
 for (int i = 0; i < n; i++)
 {
  for (int j = 0; j < n; j++)
   printf ("%d\t", dist[i][j]);
  printf("\n");
 }
 return 0;
}
```

Analysis of algorithms

OUTPUT:



TIME COMPLEXITY :

- Floyd Warshall Algorithm consists of three loops over all the nodes.
- The inner most loop consists of only constant complexity operations.
- Hence, the asymptotic complexity of Floyd Warshall algorithm is O(n3).
- Here, n is the number of nodes in the given graph.

CONCLUSION:

By using Floyd Warshall Algorithm, one can find minimum cost distance between all pair of vertices.